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ORIGINAL INVESTIGATION

Effectiveness of Accentuated Eccentric Loading: Contingent on Concentric Load

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Purpose: To identify acute effects of a single accentuated eccentric loading (AEL) repetition on subsequent back-squat kinetics and kinematics with different concentric loads. ***Methods:*** Resistance-trained men (N = 21) participated in a counterbalanced crossover design and completed 4 protocols (sets × repetitions at eccentric/concentric) as follows: AEL65, 3 × 5 at 120%/65% 1-repetition maximum (1-RM); AEL80, 3 × 3 at 120%/80% 1-RM; TRA65, 3 × 5 at 65%/65% 1-RM; and TRA80, 3 × 3 at 80%/80% 1-RM. During AEL, weight releasers disengaged from the barbell after the eccentric phase of the first repetition and remained off for the remaining repetitions. All repetitions were performed on a force plate with linear position transducers attached to the barbell, from which eccentric and concentric peak and mean velocity, force, and power were derived. ***Results:*** Eccentric peak velocity (−0.076 [0.124] $m \cdot s^{-1}$; $P = .01$), concentric peak force (187.8 [284.4] N; $P = .01$), eccentric mean power (−145.2 [62.0] W; $P = .03$), and eccentric peak power (−328.6 [93.7] W; $P < .01$) during AEL65 were significantly greater than TRA65. When collapsed across repetitions, AEL65 resulted in slower eccentric velocity and power during repetition 1 but faster eccentric and concentric velocity and power in subsequent repetitions ($P \leq .04$). When comparing AEL80 with TRA80, concentric peak force (133.8 [56.9] N; $P = .03$), eccentric mean power (−83.57 [38.0] W; $P = .04$), and eccentric peak power (−242.84 [67.3] W; $P < .01$) were enhanced. ***Conclusions:*** Including a single supramaximal eccentric phase of 120% 1-RM increased subsequent velocity and power with concentric loads of 65% 1-RM, but not 80% 1-RM. Therefore, AEL is sensitive to the magnitude of concentric loads, which requires a large relative difference to the eccentric load, and weight releasers may not need to be reloaded to induce performance enhancement.

Keywords: kinetics, kinematics, velocity, weight releasers, power

Improvements in strength and power are common resistance-training program goals, leading practitioners to seek efficient and effective methods of enhancing force and velocity. Postactivation potentiation is the process of enhancing muscle performance through selected muscle contractile activities, which require long rest periods (eg, 7–10 min) to counter the fatigue.[1] Accentuated eccentric loading (AEL),[2,3] where eccentric loads are greater in comparison to concentric loads has been shown to increase force, velocity, and power during an exercise[2,4] without the need for extended rest periods often associated with other means of post-activation potentiation.[1] This has been demonstrated using AEL during countermovement jumps, where jump height and power increased and returned to baseline quicker than traditional post-activation potentiation training.[5] Generally, AEL is implemented to potentiate concentric performance, an idea that has been supported by some[6–8] but not others,[9,10] and lacks support in exercises such as the squat.[7,10,11] By understanding the basic mechanisms behind the phenomenon, it may be possible to increase the effectiveness of AEL.

Compared with concentric muscle actions, eccentric muscle actions with that same load may result in lower motor unit recruitment and firing rates.[12] Yet, it can be expected that if the eccentric load is greater than the concentric load, the neuromuscular responses would be similar. As such, AEL can increase relative eccentric intensity resulting in increased rates of muscle lengthening, tendon extension, and motor unit activation.[2,3,13] As high rates of muscle lengthening may enhance the responses from stretch-shortening reflexes,[14,15] eccentric velocity during AEL should be investigated because it could reflect the rate of muscle lengthening. Thus, following faster eccentric velocities, one would expect a reciprocating effect on concentric performances.

However, most research on AEL has evaluated single repetitions and some have not reported eccentric velocity.[6–9] This may be problematic when exploring AEL, since heavy eccentric loads of different magnitudes can result in different volitional eccentric velocities.[16] For example, 1 investigation found greater eccentric rate of force development during a repetition with AEL, as well as 2 subsequent repetitions without accentuated loading, but concentric performances were unaffected.[10] It is possible that the eccentric movement was not fast enough or that the concentric load was too great to result in potentiation effects.[11] Yet, no study has investigated how the magnitude of concentric loads may influence the effects of AEL.

Weight releasers have become popular as a practical means to implement AEL during barbell exercises.[3,7,8,10] However, the cumbersome nature of weight releasers results in a predominant use of single repetitions in research and training,[7–9] though recently the effects of multiple repetitions have been explored.[10,11] The analysis of multiple repetitions following a repetition with AEL would provide the time course of responses and further practical evidence for whether single or multiple repetitions should be performed. Thus, the purpose of the current study was to evaluate the acute effects of supramaximal AEL on the kinetics and kinematics during back-squat training with concentric loads of different magnitudes.

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Methods

Subjects

Twenty-one resistance-trained men (age, 24.0 [4.2] y; body mass, 85.5 [11.1] kg; height, 1.75 [0.07] m; body fat percentage, 17.8 [2.3]; back-squat 1-repetition maximum (1-RM), 169.8 [29.0] kg; back squat to 1-RM ratio, 2.0 [0.3]) with a required minimum of 2 consistent years training with the back-squat exercise at least once per week participated in the study. Subjects reported 6.8 (3.0) y of structured resistance training and 1 to 3 squat training sessions per week. Further inclusion/exclusion criteria included the following: men aged 18–35 y with no lower body musculoskeletal injury in the past 6 mo, and the ability to back squat at least 1.5 times their body weight (self-reported and confirmed by 1-RM testing). Prior to data collection, the expectations and risks of the study were explained and written informed consent was obtained for all subjects. This study was conducted according to the Declaration of Helsinki guidelines and all procedures were approved by the George Mason University Institutional Review Board for use of human subjects in research.

Design

A repeated-measures crossover design was used to examine the acute effect of AEL on the kinetics and kinematics of subsequent back-squat repetitions with different concentric loads. First, anthropometric measurements, body composition, and 1-RM were assessed followed by weight releaser familiarization. After 72 h of rest, the first of 4 experimental sessions took place with all experimental sessions being separated by 48 h and taking place at the same time each day. In random order, experimental testing consisted of 2 AEL and 2 traditional (TRA) protocols that required concentric back-squat loads of 65% and 80% of 1-RM. Subjects were asked to refrain from any strenuous physical activity or lower body resistance exercise for 48 h prior to 1-RM testing until completion of the study. Subjects were also instructed to wear the same footwear, which they typically used to squat in (ie, flat heeled sneakers or weightlifting shoes) to all sessions.

1-RM and Familiarization

Subjects' height, mass, and body composition (dual-energy X-ray absorptiometry) were measured, prior to resistance-training procedures. Then subjects performed a standardized supervised warm-up, which consisted of 5 min on a stationary cycle, followed by 10-min of dynamic stretches. After completing the standardized warm-up and with an unweighted barbell, subjects demonstrated the ability to squat until the top of their thigh was parallel to the floor, and settings for safety bar height were noted. Then subjects began an incremental squat warm-up based upon the subjects' self-reported 1-RM, and included 5 repetitions at 40% 1-RM, 2-min rest, 3 repetitions at 60% 1-RM, 3-min rest, and 2 repetitions at 80% 1-RM. After a 3-min rest, single maximal attempts were made, with progressive and self-selected incremental loading, until the subject reached his 1-RM. After 2 failed attempts or the inability to continue with correct form, the 1-RM testing ended. All subjects reached their 1-RM within 5 attempts. Rest between attempts was 3 to 5 min. Squat depth, for all testing, was visually assessed by the same trained researcher (National Strength and Conditioning Association—Certified Strength and Conditioning Specialist) and supplemented with auditory signaling from a squat beeper (bigger, faster, and stronger; safety squat). The foot placement, rack height, and safety bar settings used during the 1-RM repetition were recorded for use in subsequent experimental sessions.

After 1-RM testing and following a short rest, weight releasers were added to an unweighted barbell (20 kg) for a total load of 55% 1-RM. The load used for weight releaser familiarization was similar to the maximal weight releaser load during experimental testing to demonstrate like protocols for walking out of the squat rack. Subjects then performed no more than 5 single repetitions separated by 30 s of rest. After each repetition, the bar was reracked, and the releasers were added to the barbell to familiarize subjects with using the weight releasers and walking out of the squat rack.

Experimental Sessions

After 72 h, subjects reported to the laboratory for the first experimental testing session and performed the same dynamic warm-up and incremental squat procedure (5 repetitions at 40% 1-RM, 2-min rest, 3 repetitions at 60% 1-RM, 3-min rest, and 2 repetitions at 80% 1-RM). Subjects then performed the first of the 4 separate experimental session in random order (sets × repetitions at eccentric/concentric) as follows: (1) AEL65—3 × 5 at 120%/65% 1-RM, (2) AEL80—3 × 3 at 120%/80% 1-RM, (3) TRA65—3 × 5 at 65%/65% 1-RM, and (4) TRA80—3 × 3 at 80%/80% 1-RM. Three repetitions were used during the 80% 1-RM protocols to reduce the likelihood of fatigue and maintain the potential for eccentric facilitation.[10]

During TRA, the eccentric load remained identical to the concentric action, at 65% or 80% 1-RM, for the entirety of the set. The variable resistance during the AEL protocol (120% 1-RM) was produced via weight releasers with loads of either 55% or 40% 1-RM. Each releaser made-up half the load and was placed 12 cm from the end of the barbell. Following the first eccentric movement, the weight releasers were disengaged upon ground contact, and the concentric portion of the first repetition and the concentric and eccentric actions of the set's subsequent repetitions were performed with only the prescribed load (65% or 80% 1RM). The rest between sets for all sessions was 3 min, of which subjects remained seated. Subjects were instructed to perform the concentric action of each repetition "as explosively as possible" with verbal encouragement throughout all experimental testing to help ensure maximal effort on each repetition.

Kinetic and Kinematic Data Acquisition and Analysis

Testing took place on a force plate (AccuPower; AMTI, Watertown, MA) with 4 linear position transducers (PT5A-150 Celesco; Measurement Specialties, Chatsworth, CA) attached to the barbell on the inside of both collars. The linear position transducers were mounted above in anterior, posterior, and left and right positions, forming 2 triangles when attached to the barbell, which enabled measurement of horizontal and vertical bar displacement.[17] Voltage outputs from the force plate and linear position transducers were converted into vertical ground reaction force and displacement, which were sampled at 1600 Hz using a custom-built interface box with an analog-to-digital card (NI cDAQ-9174; National Instruments Corp, Austin, TX). Initiation of movement for data analysis began when velocity or displacement increased above 0.01 $m \cdot s^{-1}$ or 0.01 m, respectively, and terminated when velocity decreased to less than 0.01 $m \cdot s^{-1}$. Prior to subjects stepping onto the force platform, calibration was completed; therefore, the force calculation included the mass of the subject

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and the load of the bar (weight of total load + total load multiplied by the acceleration of the system's center of mass). The displacement data were single and double differentiated to determine the velocity and acceleration. Power was calculated as the product of net force and velocity for each repetition of overall sets. LabView full development system (version 8.2; National Instruments, Austin, TX) was used to collect and calculate data from the devices to output kinetics and kinematics during the back squat. Mean and peak velocity and power during concentric and eccentric phases were obtained and used for comparisons. Due to the nature of AEL, the forces were only analyzed during concentric phases since the greater load of 120% 1-RM during eccentric training would yield higher eccentric forces.[8]

Statistical Analysis

Data were approximately normal, according to results from visual inspection of histograms and Shapiro–Wilk test, and no outliers were present. If the assumption of sphericity was violated, a Greenhouse–Geisser correction was performed, since epsilon values were <0.75.[18] After meeting all assumptions, analyses of variance with repeated measures were run separately for 65% and 80% 1-RM protocols with protocol (AEL and TRA) by set (3) by repetition (5 or 3) for all kinetic and kinematic variables. A Bonferroni post hoc test was used when a significant univariate effect was identified. Mean and SDs were calculated, as well as mean differences and respective 95% confidence intervals (CIs). Significance was set at $P \leq .05$. All statistical procedures were conducted using the Statistical Package for the Social Sciences SPSS (version 24; IBM Corp, Somers, NY). In addition, effect sizes were calculated using Cohen d and ranges defined according to Hopkins as trivial, $d = 0$ to 0.2; small, $d = 0.2$ to 0.6; moderate, $d = 0.6$ to 1.2; large, $d = 1.2$ to 2.0; and very large $d > 2.0$.[19]

Results

Protocol by repetition interactions for velocity, force, and power is presented in Table 1. Significant pairwise comparisons for velocity and power are indicated in Figures 1–4, along with moderate to large effect sizes. All pairwise comparisons and effect sizes for force are presented in Table 2. Protocol by set interactions and main effects is presented in the text below.

Velocity

During AEL65, eccentric mean velocity in set 3 was –0.038 m·s^{-1} (95% CI, –0.073 to –0.004; $P = .032$) faster compared with TRA65 ($F_{2,40} = 6.39$; $P < .01$). During AEL65, eccentric peak velocity in

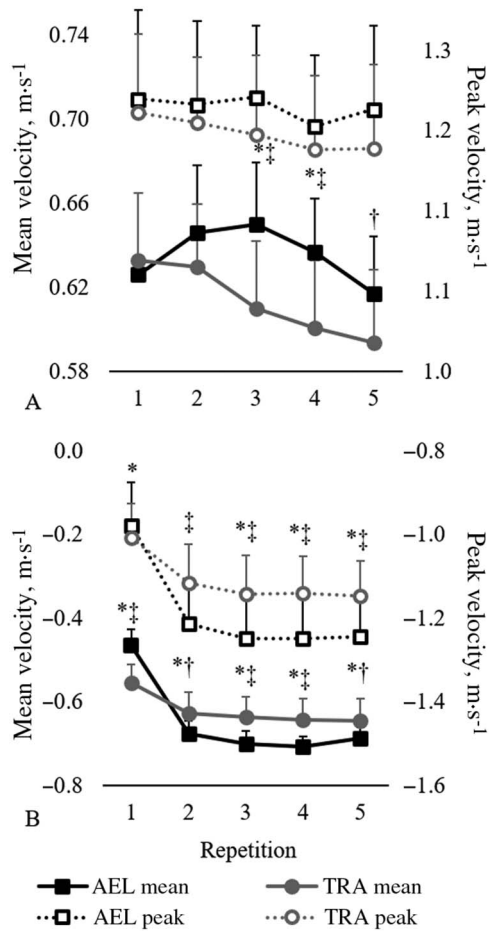

Figure 1 — Mean and peak (A) concentric and (B) eccentric velocity of each repetition collapsed across sets of 65% 1-repetition-maximum back squats. AEL indicate accentuated eccentric loading; TRA, traditional. *Significant difference between AEL and TRA sets ($P < .05$). †Moderate effects, $d = 0.6$ to 1.2. ‡Large effects, $d = 1.2$ to 2.0.

Table 1 Analysis of Variance Protocol by Repetition Interactions

	AEL65 vs TRA65	AEL80 vs TRA80
Concentric mean velocity	$F_{2.9,57.9} = 5.07$, $P < .01$; $\eta^2 = .20$	$F_{2,40} = 10.27$, $P < .01$, $\eta^2 = .34$
Eccentric mean velocity	$F_{2.4,48.0} = 26.80$, $P < .01$; $\eta^2 = .57$	$F_{2,40} = 15.27$, $P < .01$, $\eta^2 = .44$
Concentric peak velocity	$F_{4,80} = 0.38$, $P = .82$; $\eta^2 = .02$	$F_{2,40} = 0.12$, $P = .88$, $\eta^2 = .01$
Eccentric peak velocity	$F_{1.7,33.0} = 7.55$, $P < .01$; $\eta^2 = .27$	$F_{2,40} = 9.54$, $P < .01$, $\eta^2 = .32$
Concentric mean power	$F_{4,80} = 2.70$, $P = .04$; $\eta^2 = .12$	$F_{2,40} = 14.53$, $P < .01$; $\eta^2 = .42$
Eccentric mean power	$F_{4,80} = 2.75$, $P = .03$, $\eta^2 = .12$	$F_{2,40} = 0.82$, $P = .45$, $\eta^2 = .04$
Concentric peak power	$F_{4,80} = 2.44$, $P = .05$; $\eta^2 = .11$	$F_{2,40} = 0.22$, $P = .80$, $\eta^2 = .01$
Eccentric peak power	$F_{4,80} = 3.85$, $P < .01$; $\eta^2 = .16$	$F_{1.4,28} = 2.11$, $P = .15$; $\eta^2 = .10$
Concentric mean force	$F_{4,80} = 28.82$, $P < .01$; $\eta^2 = .59$	$F_{2,40} = 55.55$, $P < .01$, $\eta^2 = .75$
Concentric peak force	$F_{4,80} = 15.34$, $P < .01$; $\eta^2 = .43$	$F_{2,40} = 8.99$, $P = .01$, $\eta^2 = .31$

Abbreviations: AEL, accentuated eccentric loading; TRA, traditional; 1-RM, 1-repetition maximum. Note: AEL65 v. TRA65, AEL with 120% 1-RM compared with TRA loading during concentric loads of 65% 1-RM. AEL80 v. TRA80, AEL with 120% 1-RM compared with TRA loading during concentric loads of 80% 1-RM.

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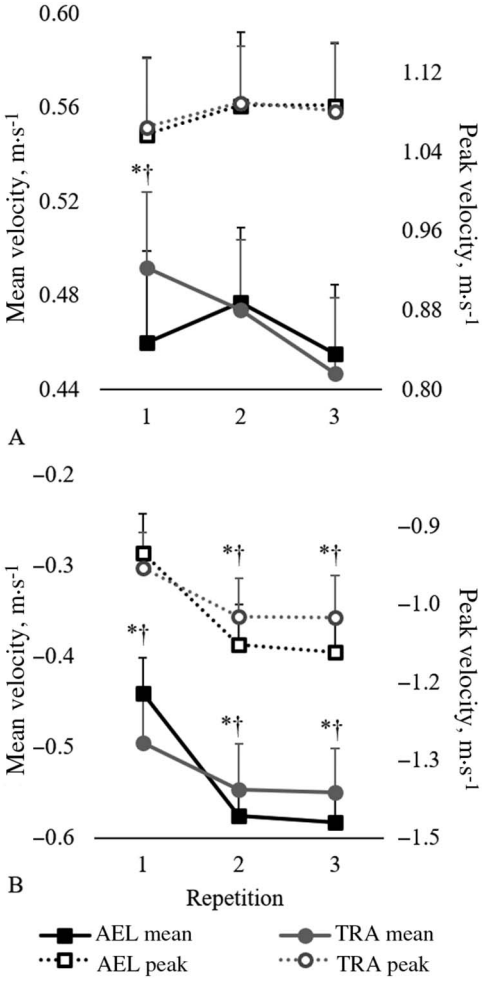

Figure 2 — Mean and peak (A) concentric and (B) eccentric velocity of each repetition collapsed across sets of 80% 1-repetition-maximum back squats. AEL indicate accentuated eccentric loading; TRA, traditional. *Significant difference between AEL and TRA sets ($P < .05$). †Moderate effects, $d = 0.6$ to 1.2.

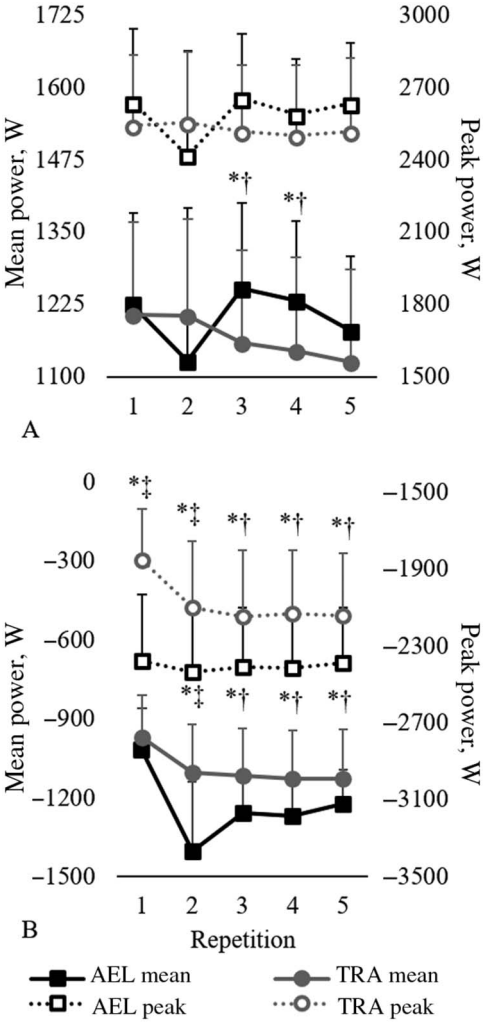

Figure 3 — Mean and peak (A) concentric and (B) eccentric power of each repetition collapsed across sets of 65% 1-repetition-maximum back squats. AEL indicate accentuated eccentric loading; TRA, traditional. *Significant difference between AEL and TRA sets ($P < .05$). †Moderate effects, $d = 0.6$ to 1.2.

set 2 and 3 was –0.095 m·s^{-1} (95% CI, –0.167 to –0.023; $P = .01$) and –0.089 m·s^{-1} (95% CI, –0.145 to –0.033; $P < .01$) faster than TRA65 ($F_{2,40} = 3.19$; $P = .05$). In the entire AEL65 protocol, eccentric peak velocity was significantly faster (–0.076 m·s^{-1}; 95% CI, –0.133 to –0.019; $F_{1,20} = 7.64$; $P = .01$). However, during 80% 1-RM protocols, no significant protocol by set interactions or main effects existed ($P > .05$).

Force

When collapsing all repetitions across each protocol, the peak concentric forces during AEL65 and AEL80 were 187.8 N (95% CI, 58.5 to 317.1; $F_{1,20} = 9.18$; $P < .01$) and 133.8 N (95% CI, 15.2 to 252.4; $F_{1,20} = 5.54$; $P = .03$) greater compared with TRA65 and TRA80, respectively.

Power

During AEL65, mean eccentric power was –145.25 W (95% CI, –273.07 to –17.42; $F_{1,20} = 5.62$; $P = .03$) and peak eccentric power was –328.60 W (95% CI, –524.11 to –113.10; $F_{1,20} = 12.29$; $P < .01$) greater than TRA65. During AEL80, mean eccentric power was –83.57 W (95% CI, –162.74 to –4.40; $F_{1,20} = 4.849$; $P = .04$) and peak eccentric power was –242.84 W (95% CI, –383.33 to –102.36; $F_{1,20} = 13.002$; $P < .01$) greater compared with TRA80.

Discussion

The primary aim of this study was to evaluate the acute effects of supramaximal AEL on kinetic and kinematic performance during the barbell back squat. Prior literature has examined how different eccentric overloads may impact subsequent performance at fixed concentric loads,[7,8] but this study is the first to evaluate how AEL affects the potentiation of different concentric loads across multiple repetitions. During repetitions with AEL in AEL65, where eccentric velocity was similar to TRA65, there was no positive or negative impact on concentric velocity. However, later repetitions where eccentric velocity during AEL65 was faster than in TRA65, the concentric velocity, force, and power during AEL65 were also superior to TRA65. Although the AEL65 and AEL80 protocols had

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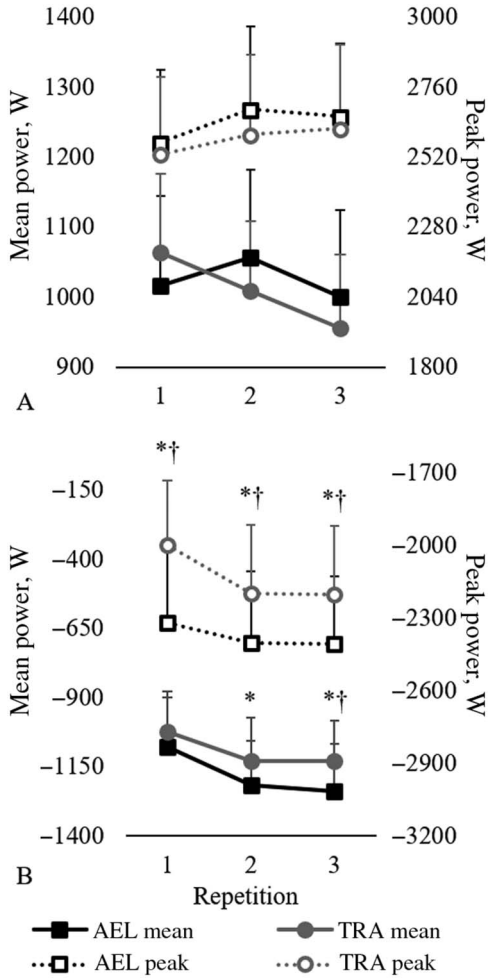

Figure 4 — Mean and peak (A) concentric and (B) eccentric power of each repetition collapsed across sets of 80% 1-repetition-maximum back squats. AEL indicate accentuated eccentric loading; TRA, traditional. *Significant difference between AEL and TRA sets ($P < .05$). †Moderate effects, $d = 0.6$ to 1.2. ‡Large effects, $d = 1.2$ to 2.0.

similar increases in eccentric velocity and power performances in later repetitions, only AEL65 resulted in greater concentric velocity after the initial repetition with eccentric overload.

It is expected that faster eccentric velocities may result in the recruitment of high threshold motor units,[20] enhanced stretch-shortening reflexes,[14] and greater muscle spindle activation,[15] which would subsequently result in greater concentric performances.[2,5] However, previous researchers, who did not report eccentric kinetics or kinematics, suggested that the stretch-shortening cycle has limited influence on the greater concentric velocity, force, and power during front squats following AEL.[7] Yet, subjects in the aforementioned study were instructed to descend for 3 s with the eccentric overload, which may likely mitigate potential benefits of the stretch-shortening cycle reflexes.[7] Others evaluating single repetitions have found no effect of AEL on velocity in subsequent jump squats with 30% 1-RM[9] or in the same repetition of bench press 1-RM.[8] However, despite subjects descending as fast as possible, eccentric velocity was not reported. Yet, others have found that even when rates of eccentric force development are faster during AEL, the concentric velocity may remain unchanged.[10]

Findings of the current study support the aforementioned lack of change, as no changes were noted in repetition 1's concentric velocity in AEL65 and peak velocity in AEL80; however, concentric mean velocity of repetition 1 was lower after AEL80 compared with TRA80. The eccentric velocity was slower during the first repetition with eccentric overload, which indicates the time to complete the eccentric phase was likely longer during AEL, unlike previous research which used equal time under tension of 3 s for both protocols.[7] Therefore, it is possible that the eccentric overload may provide further potentiation when the eccentric tempo is controlled in both protocols (limiting use of the stretch-shortening cycle), which is likely the result of the greater muscle activation from heavier eccentric loads in AEL than TRA. However, the eccentric overload during AEL may result in slower concentric velocities if compared with TRA with faster eccentric movements, since the reflexes during the stretch-shortening cycle may be reduced from slower muscle lengthening. Moreover, to control the weight releasers when descending, the pacing strategy may be affected despite intended effort to be as explosive as possible. This change in pacing strategy may have influenced the subject's ability to initiate the concentric portion of the first repetition in AEL80 and may explain why the first repetition's mean, but not peak, velocity was lower than TRA80. However, during AEL65, the concentric

Table 2 Comparison of Concentric Back-Squat Force Between AEL and TRA Loading Repetitions Collapsed Across 3 Sets

		Concentric mean force			Concentric peak force		
Load	Repetition no.	AEL	TRA	ES	AEL	TRA	ES
65% 1-RM	1	2127 (206)	1844 (206)	1.4*	2896 (249)	2507 (305)	1.4*
	2	1962 (202)	1848 (206)	0.6	2818 (295)	2597 (341)	0.7*
	3	1899 (181)	1843 (206)	0.3	2749 (300)	2619 (348)	0.4*
	4	1903 (180)	1843 (206)	0.3	2717 (310)	2623 (348)	0.3
	5	1880 (180)	1843 (205)	0.2	2741 (310)	2635 (342)	0.3
80% 1-RM	1	2308 (191)	2119 (205)	1.0*	3039 (264)	2812 (290)	0.8*
	2	2171 (175)	2107 (203)	0.3	2996 (309)	2886 (317)	0.4
	3	2165 (181)	2110 (204)	0.3	2969 (327)	2904 (325)	0.2

Abbreviations: AEL, accentuated eccentric loading; ES, effect size; TRA, traditional; 1-RM, 1-repetition maximum. Note: AEL with 120% 1-RM; ES calculated as Cohen *d*.
*Statistical significance.

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loads were lower and likely allowed the subjects to recover their pacing strategy immediately after the weight releasers disengaged, rather than later in the repetition, which was the case for AEL80.

During fast lengthening contractions, the number of cross-bridge formations may be increased due to the use of a second myosin head or rapid reattachments following forceful detachments and lead to recruitment of higher threshold motor units.[21,22] It is then expected that differences in eccentric velocities may have to reach above a threshold before significant increases in concentric performances are noted. Thus, in later repetitions where eccentric velocity was enhanced (effect size: repetition 2, −1.01; repetition 3, −1.33; repetition 4, −1.40; and repetition 5, −0.89), concentric velocity was faster during AEL65 compared with TRA65 for repetitions 3 and 4. Likewise, despite the faster eccentric velocities in AEL80 (effect size: repetition 2, −0.72 and repetition 3, −0.84), concentric velocity was unaltered in repetitions 2 and 3. This may suggest that the protocol effect on eccentric velocity has to be large to further potentiate concentric performances. Further, previous literature using 105% 1-RM for eccentric overload found concentric mean velocity at 80% 1-RM to be unaffected, despite greater eccentric rate of force development due to overload.[10] Therefore, it is possible that the eccentric overload may allow faster eccentric velocities in following repetitions, which enable greater concentric performances. However, concentric velocity and power during back squats with concentric loads of 80% 1-RM may be unaffected by the previous eccentric overload.

Although concentric forces decline as concentric velocity rises, eccentric forces may increase with eccentric velocity to a certain extent.[23] Further, the inhibitory mechanisms during eccentric muscle actions may be reduced for resistance-trained individuals[24] and thus may contribute to the force producing ability of the highly resistance-trained subjects in the current study.[25] In the current study, concentric forces were greater following AEL, specifically during the initial eccentrically overloaded repetition. Therefore, due to the inverse relationship of velocity and force in concentric actions, those repetitions in which velocity was lower produced greater ground reaction forces. However, this does not explain why concentric forces were greater when concentric velocity was also faster. The important structural component of muscle, titin, has spring like properties that produce passive forces, which may increase force producing capabilities regardless of lengthening velocity.[26,27] Another rationale for simultaneous increases in velocity and force following an AEL repetition is the potential for different pacing strategies following the release of the weight releasers. Thus, the lifter could initialize the movement with greater acceleration or accelerate for a longer portion of the concentric phase following AEL, resulting in greater peak forces in early or late phases and faster concentric velocity throughout the repetition.

Ultimately this may indicate a shift in the force–velocity curve following AEL. In agreement, 105% and 120% eccentric loads elicited greater concentric peak force during 90% 1-RM front squats.[7] However, no changes in force may occur when using loads ≤50% 1-RM,[8,9] while forces may be reduced when using maximal concentric repetitions.[8] Therefore, effects of an eccentrically overloaded repetition on ground reaction forces may be susceptible to concentric loads and no effect may exist if the concentric load is too light, while negative effects may occur during maximal repetitions. Again, this highlights that there is likely an optimal range of loads to be used during concentric actions when using AEL, similar to what prior literature has defined as optimal ranges for training power output.[28,29] However, more research is needed prior to defining what may be the best range of loads to use during AEL.

In the current study, eccentric mean and peak power were greater during nearly every repetition. Yet, mean concentric power was greater only in repetitions 3 and 4 during AEL65 and was likely driven by the enhanced concentric velocity during these repetitions. In contrast, no increases in concentric power existed during AEL80. These findings may suggest differing effects of AEL based upon concentric load selection. Others have found that higher eccentric loads of 120% elicited greater improvements in front squat power compared with using 105% for the eccentric load.[7] Therefore, although the absolute magnitude of both eccentric and concentric loads may influence the kinetic and kinematic enhancement ensued by AEL, having a large gap between the eccentric overload and subsequent concentric load may be essential.

Practical Applications

The AEL via weight releasers seems to not only be sensitive to the magnitude of eccentric loads, but also to the load used during concentric actions. Therefore, AEL seems to lose its positive effects at heavier concentric loads; however heavier concentric loads reduce the relative difference to the eccentric load, which may explain the importance of the concentric load used. Further, due to the cumbersome nature of using weight releasers, they are typically used with single repetitions or reloaded after each repetition. Overall, our findings allude to important practical considerations, as weight releasers may affect the pacing strategy when applied and impede concentric performance of that repetition and the potentiating effects seem to occur in repetitions following those with AEL. Finally, since these findings are supported by prior work,[10,11] it is possible to draw more robust conclusions and inform program development and implementation.

Conclusions

The inclusion of a single supramaximal eccentrically loaded repetition of 120% 1-RM increased subsequent velocity and power with concentric loads of 65% 1-RM, but not 80% 1-RM. Therefore, effects of supramaximal AEL on concentric velocity and power are subject to the magnitude of load used during concentric actions.

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